Theoretical Exercises – Sheet 9 - Solution

1. Recap the values for the attribute *age* from Exercise Sheet 2. The data tuples were 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70 (in increasing order).

   a. Use smoothing by bin means (use a bin of depth 3)

Bin 1	13	15	16	14.6666667		
Bin 2	16	19	20	18.3333333		
Bin 3	20	21	22	21		
Bin 4	22	25	25	24		
Bin 5	25	25	30	26.6666667		
Bin 6	33	33	35	33.6666667		
Bin 7	35	35	35	35		
Bin 8	36	40	45	40.3333333		
Bin 9	46	52	70	56		

   b. What is the effect of smoothing for the given data?

   Less impact of noise i.e. the outlier “70”

   c. What other methods are there for smoothing?

   Smoothing by bin boundaries, Regression, Filtering

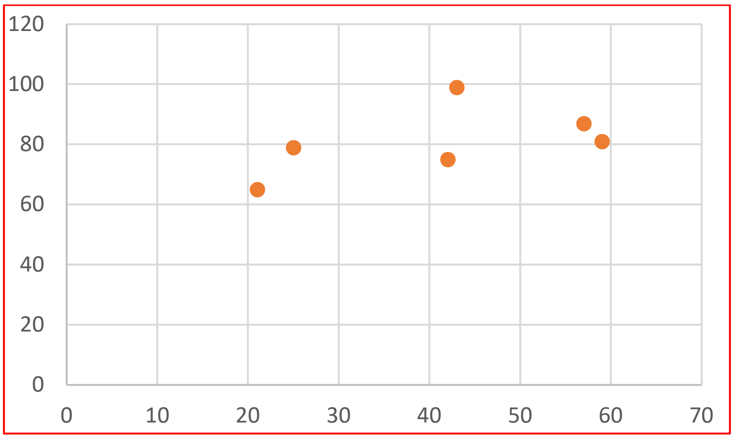
   d. How might you determine outliers in the data?

   Here: Univariate distribution, creation of a boxplot, outlier outside 1.5 x IQR

2. Suppose that a hospital tested the age and the glucose level data for 6 randomly selected adults as follows:

age	43	21	25	42	57	59
glucose	99	65	79	75	87	81

   a. Draw a scatter plot based on these two attributes.



   b. Calculate the correlation coefficient by hand and interpret the result.

   Mean x = 41.1666, Mean y = 81

SUBJECT	AGE X	GLUCOSE LEVEL Y	XY	X^2	Y^2
1	43	99	4257	1849	9801
2	21	65	1365	441	4225
3	25	79	1975	625	6241
4	42	75	3150	1764	5625
5	57	87	4959	3249	7569
6	59	81	4779	3481	6561
Σ	247	486	20485	11409	40022

$$r_{xy} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

The correlation coefficient is r=0.5298, which means the variables have a moderate positive correlation.
Positive → when age is higher, more glucose
0.5 is moderate → check scatter plot
But we could also say that our dataset is too small and the correlation coefficient is too low to make a strong conclusion.

3. Given is the following data that shows the observed number of fines in relation to the size of the car:

	Carsize	
NrOfFines	Compact	Luxury
1 or less	30	0
2 or 3	20	10
more than 3	10	30

Excerpt from a chi-square distribution table:

DF\AREA	.995	.990	.975	.950	.900	.750	.500	.250	.100	.050
1	0.00004	0.00016	0.00098	0.00393	0.01579	0.10153	0.45494	1.32330	2.70554	3.841
2	0.01003	0.02010	0.05064	0.10259	0.21072	0.57536	1.38629	2.77259	4.60517	5.991
3	0.07172	0.11483	0.21580	0.35185	0.58437	1.21253	2.36597	4.10834	6.25139	7.814

a. Calculate the marginal distributions and the expected values if *Carsize* and *NoOfFines* would be independent.

	Carsize		
NrOfFines	Compact	Luxury	
1 or less	30	0	30
2 or 3	20	10	30
more than 3	10	30	40
	60	40	100

b. Check if *Carsize* and *NrOfFines* depend on each other (use DOF = 2, α=0.1)

http://www.mathods.com/mathe.php?con=Statistik&can=Hypothesentests&cun=Chi-Quadrat-Unabh%E4ngigkeitstest

Hinweis: Voraussetzung zur Durchführbarkeit eines Chi-Quadrat-Unabhängigkeitstestes:

$$\tilde{n}_{ij} = n \cdot \hat{h}_{i\bullet} \cdot \hat{h}_{\bullet j}$$

$$\geq 5 \qquad \forall i, j$$

n_{ij} beobachtete absolute Häufigkeiten für das Wertepaar (x_i, y_j), $\tilde{n}_{ij}$ erwartete absolute Häufigkeiten für das Wertepaar (x_i, y_{ij}) bei Unabhängigkeit, $\hat{h}_{i\bullet}$ bzw. $\hat{h}_{\bullet j}$ geschätzte relative Randhäufigkeit für x_i bzw. y_j, n Anzahl der Stichprobenpaare

Somit muss man die Mittelklasse und die Luxusklasse zusammenlegen, und daraus ergibt sich:

$$\tilde{n}_{11} = n \cdot \hat{h}_{1\bullet} \cdot \hat{h}_{\bullet 1} = 100 \cdot 0,46 \cdot 0,3 = 18 \geq 5 \qquad \forall i, j$$

$$\tilde{n}_{12} = n \cdot \hat{h}_{1\bullet} \cdot \hat{h}_{\bullet 2} = 100 \cdot 0,6 \cdot 0,3 = 18 \geq 5 \qquad \forall i, j$$

$$\tilde{n}_{13} = n \cdot \hat{h}_{1\bullet} \cdot \hat{h}_{\bullet 3} = 100 \cdot 0,6 \cdot 0,4 = 24 \geq 5 \qquad \forall i, j$$

$$\tilde{n}_{21} = n \cdot \hat{h}_{2\bullet} \cdot \hat{h}_{\bullet 1} = 100 \cdot 0,4 \cdot 0,3 = 12 \geq 5 \qquad \forall i, j$$

$$\tilde{n}_{22} = n \cdot \hat{h}_{2\bullet} \cdot \hat{h}_{\bullet 2} = 100 \cdot 0,4 \cdot 0,3 = 12 \geq 5 \qquad \forall i, j$$

$$\tilde{n}_{23} = n \cdot \hat{h}_{2\bullet} \cdot \hat{h}_{\bullet 3} = 100 \cdot 0,4 \cdot 0,4 = 16 \geq 5 \qquad \forall i, j$$

Einsetzung von $p = 2$und $q = 3$:

$$= \sum_{i=1}^{2} \sum_{j=1}^{3} \frac{(n_{ij} - \tilde{n}_{ij})^2}{\tilde{n}_{ij}}$$

Einsetzung aller bekannten Werte:

$$= \frac{(30-18)^2}{18} + \frac{(20-18)^2}{18} + \frac{(10-24)^2}{24} + \frac{(0-12)^2}{12} + \frac{(10-12)^2}{12} + \frac{(30-16)^2}{16}$$

$$\approx 40,9722$$

$$\chi = \chi^2_{(p-1)(q-1);1-\alpha}$$

Einsetzung von $p = 2$, $q = 3$ und $\alpha = 0,1$:

$$= \chi^2_{(2-1)(3-1);1-0,01}$$

$$= \chi^2_{2;0,9}$$

Den Wert aus der Tabelle der Quantile der Chi-Quadrat-Verteilung ablesen:

$$\approx 4,605$$

Da

$$T \qquad > \chi$$

$$\Leftrightarrow 40,9722 > 4,605$$

ist, kann H_0 abgelehnt werden.

4. Shortly discuss issues to consider when merging data from different sources.

Formats, Typos, Different Units, Redundant Attributes, Irrelevant Features, Duplicates Sparseity (i.e. with new attributes and not matches of the tuples)

5. What are the value ranges of the following normalization methods?

   a. min-max normalization

      The range may be pre-defined by R=[minR, maxR]

   b. normalization by decimal scaling

      The range is R=[-1,1]

   c. z-score normalization

      The range R=[-∞, ∞] in terms of standard deviations

6. Remember the data for the attribute *age*. The *age* values for the data tuples are:
   13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70
   (in increasing order).

   a. Use min-max normalization to transform the value 35 for *age* onto the range [0,1] and [-1,1]

      minA = 13, maxA = 70

      v'i = (35-13)/(70-13) = 22/57 = 0.386 in R=[0,1]

      v'i = 0.386 x (maxR – minR) + minR = 0.386 x 2 -1 = -0.228 in R=[-1,1]

      41,5 wäre der Nullwert

   b. Use normalization by decimal scaling to transform the value of 35.

      ¦maxA¦ = 70 → j = 2

      v'i = 35/10^2 = 0.35

   c. Use z-score normalization to transform the value of 35, where the standard deviation of *age* is 12.94

      Mean(A) = 29.96

      v'i = 35-29.96 / 12.94 = 0.389

   d. Comment on which normalization method you would prefer to use for the given data and why.

      Min-max would make sense with 0 and 100 as extreme values for age. Z-score? Is age normal distributed?

7. Why do we need normalization what is the difference in normalization instead of just keeping the data in its base format?

   Normalization is essential to present data in a consistent state, enabling a fair comparison of the weight of different parameters. There are several key reasons why normalization is important:

   • Handling variables with different units: When variables have different units or scales, those with larger numerical ranges can dominate the algorithm. For example, consider the parameter age with a range of 0–100, and another parameter income with a range of 10,000–100,000. In this case, income could have a much larger impact on the algorithm simply because its values are numerically higher, even if it is not more important than age.

- Enabling comparison of parameter weights: Normalization allows us to compare the relative importance of parameters on an equal scale. For instance, in a model where the weight for age is 0.3 and the weight for income is 0.31, we can conclude that both parameters have a similar influence on the model. Without normalization, such comparisons may be misleading because the original scales of the parameters could distort the weights.

8. Theme park visitors have been recorded as by various criteria including their *age*:

No	1	2	3	4	5	6	7	8	9	10	11	12
Age	7	22	9	12	22	36	11	13	38	64	7	8

a. Do an equidepth binning on the attribute *Age*. When done the attribute should hold just 3 distinct values. Decide on meaningful labels for the created bins. Always specify the bins' lower and upper boundaries.

Overall there are 12 instances. When doing an equidepth binning there is the same number of instances in each bin. So we come to 12 / 3 = 4 instances in every bin. Therefore we install an interval boundary at each 4th instance. To do so we first have to order the instances by the attribute to be discretized (here "Age"):

7 7 8 9 11 12 13 22 22 36 38 64

When strictly applying the algorithm we end up with these bins (meaningful names given for the individual bins):

Bin 1: lower boundary ≥ 7 years upper boundary < 9 years proposed label: "Children"
Bin 2: lower boundary ≥ 11 years upper boundary < 22 years proposed label: "Youths"
Bin 3: lower boundary ≥ 22 years upper boundary ≤ 64 years proposed label: "Adults"

One of the two instances of age 22 is in the second bin, the other one in the third bin! Remember, the objective is to sort unknown instances. In the end determining the bin boundaries is important, not the location of the training set's instances. It has to be made sure that any unknown instance can be assigned to a bin unambiguously. Here this is ensured by defining the upper boundary of bin 2 as < 22 and the lower boundary of bin 3 as >= 22. Bins must not overlap!

There is a "gap" between instances No. 4 and 5 – because the training set is distributed that way. Unknown instances, however, might well have values inside that gap, e.g. an age of 10. Hence the upper boundary of bin 1 and the lower boundary of bin 2 have to be selected in a way that every unknown instance's value can be assigned to a bin. Bins have to be adjacent!

It is true as well that unknown instances might have values below the minimum or above the maximum of the training set. With background knowledge sensible values for the lower boundary of the first bin and the upper boundary of the last bin have to be chosen, e.g. 0 and 100 here.
On second thought the bins are slightly adapted to:

Bin 1: lower boundary ≥ 0 years upper boundary < 11 years proposed label: "Children"
Bin 2: lower boundary ≥ 11 years upper boundary < 22 years proposed label: "Youths"
Bin 3: lower boundary ≥ 22 years upper boundary ≤ 100 years proposed label: "Adults"

b. Now do an equiwidth binning on the attribute *Age* – again with 3 bins. What would be meaningful labels for the bins this time? Again specify the bins' lower and upper boundaries.

Now the value domain has to be divided into three equal parts. The domain's range is from 7 to 64 years that is 57 years. 57 / 3 = 19. So, one bin is exactly 19 years wide. Allowing for sensible minimum and maximum values the bins can be defined as:

Bin 1: lower boundary ≥ 0 years upper boundary < 26 years proposed label: "young"
Bin 2: lower boundary ≥ 26 years upper boundary < 45 years proposed label: "middle"
Bin 3: lower boundary ≥ 45 years upper boundary ≤ 100 years proposed label: "old"

c. Which discrete value for the attribute "Age" will the following unknown instances get assigned?

Age	equidepth	equiwidth	
8 years	Children	young	
10 years	Children	young	depending on handling the "gap" between bin 1 and 2 for equidepth binning
22 years	Adults	young	depending on the definition of the boundaries (including / excluding on which side?)
45 years	Adults	old	depending on the definition of the boundaries (including / excluding on which side?)
6 years	Children	young	outside of training set's value domain; most fitting discrete value is assigned
70 years	Adults	old	outside of training set's value domain; most fitting discrete value is assigned

9. Use the data from exercise 3 and sketch examples of each of the following sample techniques: SRSWOR, SRSWR and stratified sampling. Use a sample size of 5 and the strata “youth”, “middle-aged” and “senior”.

SRSWOR: 16, 20, 25, 35, 40

SRSWR: 15, 15, 22, 22, 22

Stratified:

youth = 13,15,16,16,19 (5 values)
middle-aged = 20,20,21,22,22,25,25,25,25,30,33,33,35,35,35,35,36,40,45,46 (20 values)
senior = 52,70 (2 values

youth 5/27 * 5 = 0.926
middle 20/27 * 5 = 3.7
senior 2/27 * 5 = 0.37

➔ 1 Sample aus youth und 4 aus middle-aged: 16,20,33,33,40